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Measuring the Driving Forces of Predictive Performance: Application to Credit Scoring

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Abstract. Because they play an increasingly important role in determining access to credit, credit scoring models are under growing scrutiny from banking supervisors and internal model validators. These authorities need to monitor the model performance and identify its key drivers. To facilitate this, we introduce the explainable performance (XPER) methodology to decompose a performance metric (e.g., area under the curve (AUC), R^2) into specific contributions associated with the various features of a forecasting model. XPER is theoretically grounded on Shapley values and is both model-agnostic and performance metric-agnostic. Furthermore, it can be implemented either at the model level or at the individual level. Using a novel data set of car loans, we decompose the AUC of a machine-learning model trained to forecast the default probability of loan applicants. We show that a small number of features can explain a surprisingly large part of the model performance. Notably, the features that contribute the most to the predictive performance of the model may not be the ones that contribute the most to individual forecasts (Shapley additive explanation). Finally, we show how XPER can be used to deal with heterogeneity issues and improve performance.

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Keywords: explainability • credit scoring • performance metrics • Shapley values

1. Introduction

Why is the area under the curve (AUC) of a given machine learning model equal to 0.7? Which features mainly explain this performance? What are the contributions of the different features to the mean squared error (MSE) of a regression model? To answer these questions, we develop and apply a general methodology, called explainable performance (XPER), which measures the marginal contribution of a particular feature to the predictive performance of a regression or classification model.

Being able to identify the driving forces of the performance of a predictive model is crucial in the context of credit scoring, where complex and opaque machine learning models are increasingly being used by banks and fintechs. Such decomposition is of primary importance both for banking supervisors and internal model validation teams as they need to understand why a given model is working or not, and for which types of borrowers. Furthermore, it also permits to address heterogeneity issues by identifying groups of individuals for which the features have similar effects on performance. One can then estimate group-specific models to improve overall performance and make sure the credit scoring model operates effectively for all borrowers. In this paper, we propose an application where we use XPER to decompose the AUC of a scoring model forecasting loan defaults and to identify the features with the strongest impact on model performance.

The XPER framework is based on Shapley values (Shapley 1953). Although Shapley values decompose a *payoff* among *players* in a *game*, XPER values decompose a *performance metric* among *features* in a *model*. More precisely, our method breaks down the difference between a performance metric and a benchmark value among the various features of the model. Formally, an XPER value is defined as the weighted

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Dempster-Shafer-esque?

average marginal contribution of a given feature to a performance metric, obtained in a set of coalitions of other features. For instance, evaluating the XPER value of x_1 in a three-feature model implies to assess the incremental performance due to x_1 by successively considering four subsets or coalitions of features: one coalition including no features, another coalition only x_2, another one only x_3, and a last coalition including both x_2 and x_3.

The application of the Shapley methodology in the context of explaining model performance is not trivial. Indeed, many Shapley values can be defined given the assumptions made on the model, the data, and the features excluded from the coalitions (see, for instance, Strumbelj and Kononenko 2010, Redell 2019, Kumar et al. 2020, Sundararajan and Najmi 2020, and Aas et al. 2021). In this paper, we define XPER values by considering the expectation of the performance metric with respect to the joint distribution of the target variable and of the features that are excluded from the coalition. A key advantage of this definition is that the benchmark value has a meaningful interpretation: It corresponds to the performance metric that we would obtain on a hypothetical sample in which the target variable is independent from all the features included in the model, that is, a fully mis-specified model with irrelevant features. Moreover, our method does not require to re-estimate the model (Gromping 2007a) or to pick ad hoc values for features excluded from the coalition (Israeli 2007).

XPER offers several other advantages. First, because the XPER decomposition is based on Shapley values, it is theoretically grounded, and XPER values satisfy the axioms of efficiency, symmetry, linearity, and null effects. These axioms collectively ensure equitable and consistent allocation of model performance contributions among the involved features.[1] Second, XPER is model-agnostic because it permits interpretation of the predictive performance of any econometric or machine learning model. Third, it is metric-agnostic because it can break down any performance metric: predictive accuracy (e.g., AUC, Gini, accuracy), goodness of fit (e.g., R^2), information criterion (e.g., Akaike information criterion (AIC), Bayesian information criterion (BIC)), loss function (e.g., MSE, mean absolute error (MAE), Q-like), or economic performance (e.g., profit-and-loss (P&L)). Fourth, XPER can be implemented either at the global level or at the local level. At the global level, the XPER value of a given feature measures its contribution to the performance of the model. At the local level, the XPER value of a given feature measures its contribution to the model performance that comes from a given individual. Finally, because the number of coalitions grows fast with the number of features, we propose two estimation procedures: an exact one when the number of features remains moderate and an approximated one, which is a modified version of the kernel Shapley additive explanation (SHAP) method of Lundberg and Lee (2017). We developed a package in Python to compute the XPER values and made it available at https://github.com/hi-paris/XPER.

We apply our methodology to a novel data set of auto loans provided by an international bank. We demonstrate the usefulness of XPER for decomposing the AUC and other performance metrics of a credit scoring model. Our findings reveal that a small number of features can explain a surprisingly large part of the model performance. Furthermore, the empirical analysis confirms that XPER values differ from standard feature importance metrics. More importantly, we show how to use XPER to boost model performance. To do so, we build homogeneous groups of individuals by clustering them based on their individual XPER values. By construction, within a given group, the features tend to have similar effects on performance (same sign, same intensity). We then show that estimating group-specific models yields to a higher accuracy than with a one-fits-all model.

Our paper contributes to the burgeoning literature on explainable artificial intelligence (XAI) (Murdoch et al. 2019, Gelman and Vehtari 2021, Molnar 2025). One well-known limitation of AI and machine learning methods comes from their opacity and lack of explainability. Most of these algorithms are considered black boxes in the sense that the corresponding outcomes cannot be easily explained to final users nor related to the initial features (Sun and Wang 2021). Recently, numerous explainability methods have been designed to explain black box models by measuring which features most affect its predictions (Zhao and Hastie 2021, Liu and Ročková 2023, Molnar 2025).

One of the most impactful XAI methods is the SHAP of Lundberg and Lee (2017), which distributes the predictions of a model among its features (see also Sundararajan et al. 2017, 2020; Lundberg et al. 2018; Casalicchio et al. 2019; Bowen and Ungar 2020; and Aas et al. 2021). However, explaining the predictive performance of a model (XPER) is not equivalent to explaining its forecasts (SHAP). SHAP identifies the features contributing the most to model predictions both when the model is correct and when the model is wrong. This implies that some features identified as contributors that lead the predictions may, in fact, mislead the model.

Although several other studies also decompose model performance using Shapley values (see Online Appendix A for a comparison table of the main studies), XPER distinguishes itself in three ways. First, XPER stands out for its versatility, enabling the analysis of the drivers of *any* performance metric for *any* estimated model. In the literature, most articles focus exclusively on the decomposition of a single performance metric (Owen and Prieur 2017, Redell 2019,

Power-set cardinality.

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Sutera et al. 2021, Verdinelli and Wasserman 2024). Second, XPER sets itself apart from most existing methodologies, which only focus on providing a global analysis of the features influencing model performance (Casalicchio et al. 2019, Ghorbani and Zou 2019, Covert et al. 2020, Fryer et al. 2021, Moehle et al. 2021, Borup et al. 2022, Zhang et al. 2023), by also enabling a local analysis. This allows us to address heterogeneity issues by identifying groups of individuals for which features exhibit similar effects on model performance. Third, XPER meaningfully decomposes the performance metric without relying on strong assumptions or re-estimating the model. In particular, many alternative approaches require re-estimating the model with different subsets of features, which may lead to omitted variable bias (Lipovetsky and Conklin 2001, Israeli 2007, Bell et al. 2024).

Beyond the algorithmic approaches discussed above, the impact of features on model performance can also be assessed within a statistically grounded hypothesis testing framework (Candes et al. 2018). For instance, Lei et al. (2018) construct finite-sample tests based on the decrease in model accuracy induced by removing features. Horel and Giesecke (2022) develop a computationally feasible significance testing procedure for the features of a predictive model. It generates a model-free measure of feature importance and also permits to detect significant feature interactions. More recently, Giesecke et al. (2025) propose a uniformly most powerful statistical test that yields exact p-values for assessing feature significance. Their method further provides confidence intervals with exact coverage for estimating population-level feature importance. It requires minimal assumptions, avoids model retraining, and remains computationally efficient even in high-dimensional settings.

2. Primer on XPER

In this section, we introduce XPER using a simple classification problem with $y \in \{0,1\}$ the target variable and three features $\{x_1, x_2, x_3\}$. We estimate a model $\hat{f}(x_1, x_2, x_3)$ on a training sample and evaluate its predictive performance by considering its AUC on a test sample. If none of the three features provides any useful information to predict the target variable, the AUC is 0.5, indicating random classification. We refer to this value as the benchmark AUC, denoted ϕ_0. In contrast, when AUC >0.5, at least one feature contains some relevant information to predict the target variable.

What is the relative contribution of each feature to this predictive performance? We answer this question using XPER by breaking down the difference between the AUC obtained on the test sample and the benchmark value, among the features of the model. Let us denote by ϕ_j the XPER contribution of feature x_j to the predictive performance of the model. For instance, the AUC of the model (e.g., predicting/detecting loan default, customer churn, fraud, money laundering) is 0.78 and its XPER decomposition is as follows:

	AUC		ϕ_0		ϕ_1		ϕ_2		ϕ_3
Test sample	0.78	=	0.50	+	0.14	+	0.10	+	0.04

It shows that the feature x_1 is the main driver of the predictive performance of the model because it explains half of the difference between the AUC of the model and its benchmark $(= 0.14/(0.78 - 0.50))$. The second most informative feature is x_2 because it explains another 35.7% $(= 0.10/(0.78 - 0.50))$ of the difference. Finally, x_3 explains the remaining 14.3%.

Such performance decomposition can prove handy in several contexts. XPER can help to highlight a potential *heterogeneity* in the predictive performance of a model. Indeed, one can compute the XPER decomposition of a given performance measure on a subset of the test sample. For instance, let us consider two subsets of the test sample, denoted A and B. The performance of the model can be evaluated and decomposed with XPER independently in each subset. The illustrative results displayed in the table below indicate that the model performs poorly in A. In this case, XPER values can be used to explain the sources of such underperformance of the model. For instance, if the XPER decomposition of the AUC over the subsets A and B is as follows, the weak performance observed in A is only due to feature x_1:

	AUC		ϕ_0		ϕ_1		ϕ_2		ϕ_3
Subset A	0.65	=	0.50	+	0.01	+	0.10	+	0.04
Subset B	0.85	=	0.50	+	0.21	+	0.10	+	0.04

XPER can also help to understand the origin of *overfitting*. Overfitting occurs when a model performs significantly better on the training sample than it does on the test sample. XPER can identify some features which contribute more to the performance of the model in the training sample than in the test sample, and thus explains overfitting. Below, the drop in AUC between the training sample (0.90) and the test sample (0.78) illustrates a typical overfitting issue. In this case, the overfitting problem is entirely due to x_2.

	AUC		ϕ_0		ϕ_1		ϕ_2		ϕ_3
Training	0.90	=	0.50	+	0.15	+	0.20	+	0.05
Test	0.78	=	0.50	+	0.15	+	0.08	+	0.05
Training – Test	0.12	=	0	+	0	+	0.12	+	0

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A related application of XPER could prove useful during the *variable selection phase* of model development. Indeed, by contrasting XPER values estimated in the training and validation sets, one could only keep the variables maintaining a high enough level of importance in the validation test.

XPER can be applied to any statistical performance metrics (e.g., R^2, Brier score, Gini index), but also to any *economic performance metrics* (e.g., P&L, return-on-investment, performance of a financial portfolio). As an example, we can decompose the P&L of a credit scoring model:

$$P\&L = \sum_{i=1}^{n}(1-\hat{y}_i)(1-y_i)\times profit + (1-\hat{y}_i)y_i \times loss,$$

where *profit* is the money made on any reimbursed loan ($y_i = 0$), and *loss* is the money lost on any defaulted loan ($y_i = 1$). In this case, XPER values would be expressed in dollar terms and would sum to the total P&L.

3. Framework and Performance Metrics

We consider a classification or regression problem involving a target variable, denoted y, taking values in $\mathcal{Y} = \{0,1\}$ or $\mathcal{Y} \subseteq \mathbb{R}$ given the problem considered. The q-vector $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^q$ refers to the input features. We denote by $f : \mathbf{x} \to \hat{y}$ an econometric model or a machine learning algorithm, where $\hat{y} \in \mathcal{Y}$ is either a classification output or a regression output, such as $\hat{y} = f(\mathbf{x})$. We impose no constraint on the model: It may be parametric or not, linear or not, a weak learner or an ensemble method, and so on. For simplicity, for parametric models, we exclude the parameters from the notation, that is, $f(\mathbf{x}) \equiv f(\mathbf{x};\theta)$. The model is estimated (parametric model) or trained (machine learning algorithm) once for all on a training sample $S_T = \{\mathbf{x}_j, y_j\}_{j=1}^{T}$. The sample size T is considered fixed. The trained model, denoted $\hat{f}(.)$, is evaluated on a test sample $S_n = \{\mathbf{x}_i, y_i, \hat{f}(\mathbf{x}_i)\}_{i=1}^{n}$ with a performance metric (PM). A PM is defined as a measure used to assess the predictive performance of a model. It quantifies the accuracy and effectiveness of a model's predictions. Standard PM for classification models include accuracy, precision, recall, F1-score, and AUC, among others. Similarly, for regression models, commonly used PM include MSE, root mean squared error (RMSE), MAE, and R^2. More generally, any information criteria, loss function, or economic performance indicator can be considered PM. See Table A2 in Online Appendix B for examples of PM.

Definition 1. A sample performance metric PM $\in \Theta \subseteq \mathbb{R}$ associated with the model $\hat{f}(.)$ and the test sample S_n is defined as $\text{PM} = \tilde{G}_n(y_1,\ldots,y_n;\hat{f}(\mathbf{x}_1),\ldots,\hat{f}(\mathbf{x_n})) = G_n(\mathbf{y};\mathbf{X})$, where $\mathbf{y} = (y_1,\ldots,y_n)'$ and $\mathbf{X} = (\mathbf{x}_1,..,\mathbf{x}_n)'$.

For instance, for a linear regression model $\hat{f}(\mathbf{x}) = \hat{\beta}_0 + \hat{\beta}_1 x_1$ and the PM defined as the opposite of the MSE, we have $G_n(\mathbf{y};\mathbf{X}) = -\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i,1})^2$. We introduce the following three assumptions on the PM.

Assumption 1. *The sample PM increases over Θ with the predictive performance.*

Assumption 1 simplifies the interpretation of the PM by asserting that higher values indicate more accurate model predictions. For example, because a higher R^2 suggests smaller differences between model predictions and target values, R^2 can be directly used as a PM. Conversely, if we want to decompose the MSE, we recommend considering the opposite of the MSE and defining PM = –MSE. Hence, higher PM values signify smaller prediction-target differences. In this case, a positive XPER value ϕ_j always indicates that the variable x_j enhances the model predictive performance.

Assumption 2. *The sample PM satisfies the following additive assumption:*

$$G_n(\mathbf{y};\mathbf{X}) = \frac{1}{n}\sum_{i=1}^{n} G(y_i;\mathbf{x}_i;\hat{\delta}_n), \tag{1}$$

where $G(y_i;\mathbf{x}_i;\hat{\delta}_n)$ denotes an individual contribution to the PM, and $\hat{\delta}_n$ is a nuisance parameter which depends on the test sample S_n.

For simplicity, we only consider models for which the outcome y_i for instance i only depends on features $\mathbf{x}_i = (x_{i,1},\ldots,x_{i,q})'$. For regression or classification models with cross-sectional interactions (e.g., spatial econometrics model) or time-series dependence, notations have to be adjusted such that $\hat{y}_i = \hat{f}(\mathbf{w}_i)$, where $\mathbf{x}_i \subseteq \mathbf{w}_i$, $\exists j \neq i : \mathbf{x}_j \subseteq \mathbf{w}_i$ and/or $y_j \subseteq \mathbf{w}_i$. Then, the additive assumption becomes $G_n(\mathbf{y};\mathbf{X}) = n^{-1}\sum_{i=1}^{n} G(y_i;\mathbf{w}_i;\hat{\delta}_n)$.

Assumption 3. *The sample PM, denoted $G_n(\mathbf{y};\mathbf{X})$, converges to the population PM, denoted $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$, where $\mathbb{E}_{y,\mathbf{x}}(.)$ refers to the expected value with respect to the joint distribution of y and $\mathbf{x}$, and $\delta_0 = \text{plim}\,\hat{\delta}_n$. In addition, $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ exists and is finite.*

These three assumptions are consistent with a wide range of PMs. In Online Appendix B, we detail $G(y;\mathbf{x};\delta_0)$ for standard PMs associated with regression or classification models. For instance, in the case of a linear regression model $\hat{f}(\mathbf{x}) = \mathbf{x}\hat{\beta}$, when the R^2 is used as PM, we have

$$R^2 = G_n(\mathbf{y};\mathbf{X}) = \frac{1}{n}\sum_{i=1}^{n} G(y_i;\mathbf{x}_i;\hat{\delta}_n) = 1 - \frac{\sum_{i=1}^{n}(y_i - \mathbf{x}_i\hat{\beta})^2}{\sum_{j=1}^{n}(y_j - \bar{y})^2}, \tag{2}$$

with $G(y_i;\mathbf{x}_i;\hat{\delta}_n) = 1 - \hat{\delta}_n^{-1}(y_i - \mathbf{x}_i\hat{\beta})^2$ and $\hat{\delta}_n = n^{-1}\sum_{j=1}^{n}(y_j - \bar{y})^2$. The corresponding population R^2 is defined

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as $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0)) = 1 - \frac{1}{\sigma_y^2}\mathbb{E}_{y,\mathbf{x}}((y-\mathbf{x}\hat{\beta})^2)$, with $G(y;\mathbf{x};\delta_0) = 1 - \delta_0^{-1}(y-\mathbf{x}\hat{\beta})^2$ and $\delta_0 = \sigma_y^2$ the variance of the target variable.

This set of assumptions imposes minimal constraints. In particular, Assumption 3 is satisfied under three key conditions related to the performance metric, the sampling assumptions, and the consistency of the estimator $\hat{\delta}_n$. First, regarding the performance metric, the function $G(\cdot)$ needs to be integrable with respect to the joint distribution of y and $\mathbf{x}$. This ensures that $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ exists and is finite. Second, depending on the type of data (cross-sectional or time series), specific assumptions about the joint distribution of y and $\mathbf{x}$ are necessary. For cross-sectional data, we assume that the elements $G(y_i;\mathbf{x}_i;\hat{\delta}_n)$ are independently distributed or weakly dependent. This assumption is typically mild because $G(y_i;\mathbf{x}_i;\hat{\delta}_n)$ generally represents a function of the error term. For example, for the (opposite) MSE, we have $G(y_i;\mathbf{x}_i;\hat{\delta}_n) = -(y_i - \hat{f}(\mathbf{x}_i))^2$. This ensures the conditions for applying the weak law of large numbers when n is large. For time-series data, we assume that the elements $G(y_i;\mathbf{x}_i;\hat{\delta}_n)$ are stationary and weakly dependent. Finally, the estimator $\hat{\delta}_n$ should be consistent for δ_0. This means $\hat{\delta}_n$ converges in probability to δ_0 as the sample size n increases.

4. XPER Values

4.1. Definition

Our objective is to identify the contribution of the model's features to its predictive performance, as evaluated by a PM on a given sample. We measure this contribution through Shapley values (Shapley 1953), a method used in game theory to fairly distribute a payoff $Val(x_1,\ldots,x_q)$ among several players $x_1,\ldots,x_q$. The Shapley value ϕ_j measures the marginal impact of a player x_j on the payoff by assessing the changes in $Val(x_1,\ldots,x_q)$ when this player is added or not to a coalition of players already in the game. Denote by $\mathbf{x}^S$ the vector of players included in a coalition S and $\mathbf{x}^{\bar{S}}$ the vector of excluded players, such that $\{\mathbf{x}\} = \{\mathbf{x}^S\} \cup \{\mathbf{x}^{\bar{S}}\} \cup \{x_j\}$ and $\mathbf{x} = (x_1,\ldots,x_q)$. The Shapley value is then defined as the weighted average of the marginal contributions associated with all possible coalitions S.

Definition 2 (Shapley 1953). The Shapley value contribution of player x_j to the payoff is

$$\phi_j = \sum_{S\in\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})} \omega_S[Val(\mathbf{x}^S \cup \{x_j\}) - Val(\mathbf{x}^S)], \tag{3}$$

$$\omega_S = \frac{|S|!(q-|S|-1)!}{q!}, \tag{4}$$

with $Val(.)$ the payoff, S a coalition of players, excluding the player of interest x_j, $|S|$ the number of players in the coalition, and $\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})$ the powerset of the set $\{\mathbf{x}\}\setminus\{x_j\}$.

By analogy, we decompose the performance metric $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ (the "payoff") among the features $x_1,\ldots,x_q$ (the "players") of the model $\hat{f}(\mathbf{x})$ (the "game"). The main difference with the previous notations is that the payoff $Val(\mathbf{x};y) = \mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ depends on the joint distribution of the features $x_1,\ldots,x_q$ and the target variable y. Formally, we define XPER as the Shapley value associated with the performance metric $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ and to the model $\hat{f}(\mathbf{x})$.

Definition 3 (XPER). The XPER value associated with the feature x_j is defined as

$$\phi_j = \sum_{S\in\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})} \omega_S \Bigg[\underbrace{\mathbb{E}_{y,x_j,\mathbf{x}^S}}_{\text{averaging}} \underbrace{\mathbb{E}_{\mathbf{x}^{\bar{S}}}}_{\text{marginalization}} (G(y;\mathbf{x};\delta_0)) - \underbrace{\mathbb{E}_{y,\mathbf{x}^S}}_{\text{averaging}} \underbrace{\mathbb{E}_{x_j\mathbf{x}^{\bar{S}}}}_{\text{marginalization}} (G(y;\mathbf{x};\delta_0)) \Bigg],$$

with S a coalition of features, excluding the feature of interest x_j, $\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})$ the powerset of the set $\{\mathbf{x}\}\setminus\{x_j\}$, and ω_S the coalition weight. The expectations are taken over the joint distribution of the features and/or target variable specified in the subscripts.

The XPER value ϕ_j measures the weighted average marginal contribution of the feature x_j to the PM over all features coalitions. For each coalition, the marginal contribution of x_j is defined as the difference between the expected values of (1) the PM obtained while including this feature in the coalition and (2) the PM obtained while excluding this feature from the coalition. In Definition 3, the term $\mathbb{E}_{\mathbf{x}^{\bar{S}}}$ refers to the marginalization with respect to the *joint distribution* of the features that are excluded from the coalition S.[2] The second expectation $\mathbb{E}_{y,x_j,\mathbf{x}^S}(G(y;\mathbf{x};\delta_0))$ refers to an averaging effect, that is, expectation with respect to the *joint distribution* of the features $\mathbf{x}^S$ included in the coalition with x_j, and the target variable y.

As an illustration, we consider a three-feature model $\hat{f}(x_1,x_2,x_3)$ and pick x_1 as the feature of interest. In Table 1, we report all the coalitions among the set $\{x_2,x_3\}$ (column 1), the associated weights computed according to Equation (4) (column 2), and the marginal contributions (column 3) used to compute the XPER value ϕ_1. For instance, when considering a coalition with only x_1, we first compute the expectation of the PM with respect to the *joint distribution* of excluded features x_2 and x_3, that is, $\mathbb{E}_{x_2,x_3}(\tilde{G}(y,\hat{f}(x_1,x_2,x_3)))$. This corresponds to a marginalization of the PM with respect to the features that are excluded from the coalition. Then, we consider the expectation of the PM with respect to the *joint distribution* of the features included

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Table 1. Components of ϕ_1 in a Three-Feature Model

S	ω_S	$\mathbb{E}_{y,x_1,\mathbf{x}^S}\mathbb{E}_{\mathbf{x}^{\bar{S}}}(G(y;\mathbf{x};\delta_0)) - \mathbb{E}_{y,\mathbf{x}^S}\mathbb{E}_{x_1,\mathbf{x}^{\bar{S}}}(G(y;\mathbf{x};\delta_0))$
$\{\emptyset\}$	1/3	$\mathbb{E}_{y,x_1}\mathbb{E}_{x_2,x_3}(G(y;\mathbf{x};\delta_0)) - \mathbb{E}_y\mathbb{E}_{x_1,x_2,x_3}(G(y;\mathbf{x};\delta_0))$
$\{x_2\}$	1/6	$\mathbb{E}_{y,x_1,x_2}\mathbb{E}_{x_3}(G(y;\mathbf{x};\delta_0)) - \mathbb{E}_{y,x_2}\mathbb{E}_{x_1,x_3}(G(y;\mathbf{x};\delta_0))$
$\{x_3\}$	1/6	$\mathbb{E}_{y,x_1,x_3}\mathbb{E}_{x_2}(G(y;\mathbf{x};\delta_0)) - \mathbb{E}_{y,x_3}\mathbb{E}_{x_1,x_2}(G(y;\mathbf{x};\delta_0))$
$\{x_2,x_3\}$	1/3	$\mathbb{E}_{y,x_1,x_2,x_3}(G(y;\mathbf{x};\delta_0)) - \mathbb{E}_{y,x_2,x_3}\mathbb{E}_{x_1}(G(y;\mathbf{x};\delta_0))$

Note. This table provides details about the XPER value computation, that is, the coalitions (column 1), the associated weights according to Equation (4) (column 2), and the marginal contributions (column 3).

in the coalition and the target, that is, y and x_1 in our example. Thus, we get an expected value $\mathbb{E}_{y,x_1}\mathbb{E}_{x_2,x_3}(\tilde{G}(y,\hat{f}(x_1,x_2,x_3)))$, where the first expectation refers to the averaging effect, whereas the second one refers to the marginalization effect. The Shapley value is then computed by summing the weighted differences in the expected PM obtained with or without the feature of interest, for all the coalitions of other features.

One advantage of our marginalization-based approach is that there is no need to re-estimate any submodel for each subset of features. We consider the expected value of the PM with respect to the features $\mathbf{x}^{\bar{S}}$ excluded from the coalitions while leaving the model $\hat{f}(\mathbf{x})$ unchanged. An alternative solution would consist in estimating a submodel for each subset of features. For example, for a model with three features x_1, x_2, x_3, the computation of the Shapley value associated with x_1 would require to estimate four submodels, namely without any feature, with x_2 only, with x_3 only, and with x_2, x_3, and then to re-estimate the same submodels while including x_1 as an additional feature. This approach was adopted, for instance, by Israeli (2007) to decompose the R^2 of a linear model. However, re-estimating a submodel with only a subset of the initial features can lead to model specification errors, for example, omitted variable bias in the case of linear regression. This specification error necessarily distorts the Shapley value and thus may lead to an unreliable decomposition of the performance metric.

XPER is a weighted average of the expected performance improvement resulting from adding the feature of interest to a model (coalition) that "turns off" this feature. Therefore, the choice of the weighting scheme is crucial. To compute the XPER values, we utilize the coalition weights derived from Shapley's original definition (Shapley 1953), as defined in Equation (4). Given this definition, the weights of coalitions are typically uniform and based on the number of features. This choice offers a significant advantage: It guarantees compliance with the four axioms that define a Shapley value—efficiency, symmetry, linearity, and null effects. Therefore, most XAI methods relying on Shapley values adopt these coalition weights (Sundararajan et al. 2017, 2020; Lundberg et al. 2018; Casalicchio et al. 2019; Bowen and Ungar 2020). For instance, Lundberg and Lee (2017) follow this approach when introducing the Shapley value to justify their SHAP method.

4.2. Axioms

The XPER values satisfy different axioms associated with Shapley values. These axioms are particularly relevant in the context of statistical performance analysis.

Axiom 1 (Efficiency). *The sum of the XPER values ϕ_j, $\forall j = 1,\ldots,q$, is equal to the difference between the performance metric $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ and a benchmark ϕ_0 such as*

$$\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0)) = \phi_0 + \sum_{j=1}^{q}\phi_j, \tag{5}$$

where $\phi_0 = \mathbb{E}_{\mathbf{x}}\mathbb{E}_y(G(y;\mathbf{x};\delta_0))$ corresponds to the performance metric associated with a population where the target variable is independent from all features considered in the model.

One of the main advantages of the XPER decomposition is that the benchmark value ϕ_0 has an insightful interpretation: It corresponds to the PM that we would obtain on a hypothetical sample in which the target variable y is independent from all model features $\mathbf{x}$, that is, in a case where the model $\hat{f}(\mathbf{x})$ is fully mis-specified. As an example, for the AUC, the benchmark ϕ_0 corresponds to the AUC associated with a random predictor and is equal to 0.5. For the sensitivity (true-positive rate), the benchmark corresponds to the probability $\Pr(\hat{y} = 1)$; for the specificity (true-negative rate), the benchmark is $\Pr(\hat{y} = 0)$, and so on. Thus, we can decompose any PM into two parts: (i) a base value ϕ_0 obtained in a hypothetical case where y and $\mathbf{x}$ would be independent, and (ii) a component determined by the XPER feature contributions, which depends on their dependence with the target, that is, their relevance:

$$\underbrace{\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0)}_{\text{population performance metric}} = \underbrace{\mathbb{E}_y\mathbb{E}_{\mathbf{x}}(G(y;\mathbf{x};\delta_0)}_{\text{expected value under independence}} + \underbrace{\sum_{j=1}^{q}\phi_j}_{\text{feature contributions}}.$$

The XPER values also satisfy the other main axioms of the Shapley values.

Axiom 2 (Symmetry). *If for all subsets $S \in \mathcal{P}(\{\mathbf{x}\} \setminus \{x_j, x_k\})$*

$$\mathbb{E}_{y,x_j,\mathbf{x}^S}\mathbb{E}_{\mathbf{x}^{\bar{S}}}(G(y;\mathbf{x},x_j;\delta_0)) = \mathbb{E}_{y,x_k,\mathbf{x}^S}\mathbb{E}_{\mathbf{x}^{\bar{S}}}(G(y;\mathbf{x},x_k;\delta_0)),$$

then $\phi_j = \phi_k$. It means that if two features x_j and x_k contribute equally to the performance of the model then their effects must be the same.

Axiom 3 (Linearity). *For a performance metric PM such that $PM = PM_1 + PM_2$, then*

$$\phi_j(PM) = \phi_j(PM_1) + \phi_j(PM_2),$$

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where $\phi_j(PM)$ refers to the XPER value of feature x_j measuring its effect on the performance metric defined as the sum of PM_1 and PM_2.

Axiom 4 (Null Effects). *If for all subsets $S \in \mathcal{P}(\{\mathbf{x}\} \setminus \{x_j\})$*

$$\mathbb{E}_{y,x_j,\mathbf{x}^S}\mathbb{E}_{\mathbf{x}^{\bar{S}}}(G(y;\mathbf{x},x_j;\delta_0)) = \mathbb{E}_{y,\mathbf{x}^S}\mathbb{E}_{x_j,\mathbf{x}^{\bar{S}}}(G(y;\mathbf{x},x_j;\delta_0)),$$

then $\phi_j = 0$. A feature without any impact on the performance of the model $\mathbb{E}_{y,\mathbf{x}}(G(y;\mathbf{x};\delta_0))$ has an XPER value equal to zero.

To illustrate these axioms, consider a linear regression model $\hat{f}(\mathbf{x}_i) = \sum_{j=1}^{q}\hat{\beta}_j x_{i,j}$ estimated on a training set, and the R^2 as PM. For simplicity, we assume that the data generating process (DGP) of the test sample $S_n = \{\mathbf{x}_i, y_i, \hat{f}(\mathbf{x}_i)\}_{i=1}^n$ satisfies $\mathbb{E}(\mathbf{x}) = \mu_q$ and $\mathbb{V}(\mathbf{x}) = \Sigma$ a positive semidefinite matrix, with $\Sigma_{k,j} = \sigma_{x_k,x_j}$ the covariance between feature x_k and x_j. Then, the XPER contribution ϕ_j of feature x_j to the R^2 is

$$\phi_j = \frac{2\hat{\beta}_j\sigma_{y,x_j}}{\sigma_y^2}, \qquad \forall j = 1,\ldots,q, \tag{6}$$

with σ_y^2 the variance of the target variable and σ_{y,x_j} its covariance with feature x_j. See Online Appendix K.1 for the proof. Formally, ϕ_j depends on the estimated parameter $\hat{\beta}_j$ (training set) and the covariance (test set) between x_j and the target variable y, that is, σ_{y,x_j}. If the DGP of the training and test samples are identical, model parameters $\hat{\beta}_j$ and covariance σ_{y,x_j} have the same sign, which means $\phi_j > 0$.[3] Otherwise, XPER values may be negative. Similarly, if $\hat{\beta}_j = 0$ (i.e., the variable is useless in the model) or if the feature is uncorrelated with the target variable on the test sample ($\sigma_{y,x_j} = 0$), then $\phi_j = 0$. Finally, a feature x_j has a larger contribution to the R^2 than a feature x_s if $\hat{\beta}_j\sigma_{y,x_j} > \hat{\beta}_s\sigma_{y,x_s}$, meaning that x_j is more related to the target variable than x_s both in-sample (through $\hat{\beta}_j$) and out-of-sample (through σ_{y,x_s}).

In this simple framework, the XPER values correspond to the contribution of the features to the explained variance, as typically computed in econometrics (Lindeman et al. 1980, Pratt 1987, Gromping 2007b). For ease of explanation, consider the following DGP:

$$y = x_1 + 3x_2 + \varepsilon, \quad \mathbb{V}(x_1) = \mathbb{V}(x_2) = 1, \quad \mathbb{V}(\varepsilon) = 10. \tag{7}$$

In this framework, the variance explained by the model is typically decomposed as follows:

$$R^2 = \frac{\mathbb{V}(\hat{y})}{\mathbb{V}(y)} = \frac{\hat{\beta}_1\sigma_{y,x_1}}{\sigma_y^2} + \frac{\hat{\beta}_2\sigma_{y,x_2}}{\sigma_y^2} = \tilde{\phi}_1 + \tilde{\phi}_2. \tag{8}$$

Thus, in our example, the R^2 is equal to 50%, with the contributions of features x_1 and x_2 to the R^2 amounting to 5% and 45%, respectively. Similarly, the XPER decomposition is

$$R^2 = \phi_0 + \frac{2\hat{\beta}_1\sigma_{y,x_1}}{\sigma_y^2} + \frac{2\hat{\beta}_2\sigma_{y,x_2}}{\sigma_y^2} = \phi_0 + \phi_1 + \phi_2, \tag{9}$$

where ϕ_j represents the XPER value associated with the feature x_j and $\phi_0 = -R^2$ the benchmark value.[4] From Equations (8) and (9), we can see that the contribution $\tilde{\phi}_j$ of the feature x_j to the explained variance can be derived from the XPER value by normalizing it with respect to the difference $R^2 - \phi_0$:

$$\tilde{\phi}_j = \frac{\phi_j}{R^2 - \phi_0}. \tag{10}$$

We generally report normalized XPER values, defined as $\phi_j/(\mathrm{PM} - \phi_0)$, where PM denotes the performance metric under consideration (e.g., R^2). Thus, in this particular setting, the XPER values exactly match the features' contributions to the explained variance.

In many cases, analytical expressions for the XPER values ϕ_j are not available. Indeed, the function $G(y;\mathbf{x};\delta_0)$ is in general nonlinear in y and $\mathbf{x}$, as the model $\hat{f}(.)$ can be highly nonlinear in $\mathbf{x}$ (e.g., machine learning model) and/or the performance metric can be nonlinear in y and $\mathbf{x}$. See Online Appendix D for the XPER values decomposition of several regression and classification performance metrics.

4.3. Individual XPER Values

4.3.1. Definition. The XPER framework can be used to conduct a global analysis of the model predictive performance through feature contributions ϕ_j, or a local analysis at the individual level.

Definition 4 (Individual XPER). The individual XPER value $\phi_{i,j}$ associated with individual i and model feature j satisfies $\phi_j = \mathbb{E}_{y,\mathbf{x}}(\phi_{i,j}(y_i;\mathbf{x}_i))$, where the random variable $\phi_{i,j}(y_i;\mathbf{x}_i)$ corresponds to individual i and feature j contribution to the performance metric defined as

$$\phi_{i,j}(y_i;\mathbf{x}_i) = \sum_{S\in\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})} w_S[\mathbb{E}_{\mathbf{x}^{\bar{S}}}(G(y_i;x_{i,j},\mathbf{x}_i^S,\mathbf{x}^{\bar{S}};\delta_0)) - \mathbb{E}_{x_j,\mathbf{x}^{\bar{S}}}(G(y_i;x_{i,j},\mathbf{x}_i^S,\mathbf{x}^{\bar{S}};\delta_0))],$$

with $\mathbb{E}_{\mathbf{x}^{\bar{S}}}(\cdot)$ the expectation with respect to the joint distribution of the features $\mathbf{x}^{\bar{S}}$, and $\mathbb{E}_{x_j,\mathbf{x}^{\bar{S}}}(\cdot)$ the expectation with respected to the joint distribution of the features $\mathbf{x}^{\bar{S}}$ and x_j.

The individual XPER value can be interpreted as the contribution of individual i and model feature j to the performance metric. For a given realization $(y_i,\mathbf{x}_i)$, the corresponding individual contribution to the

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performance metric can be broken down into

$$G(y_i;\mathbf{x}_i;\delta_0) = \phi_{i,0} + \sum_{j=1}^{q} \phi_{i,j}, \tag{11}$$

where $\phi_{i,j}$ is the realization of $\phi_{i,j}(y_i;\mathbf{x}_i)$, and $\phi_{i,0}$ is the realization of $\phi_{i,0}(y_i) = \mathbb{E}_{\mathbf{x}}(G(y_i;\mathbf{x};\delta_0))$. The benchmark $\phi_{i,0}$ corresponds to the contribution of an individual to the performance metric when the target variable y_i is independent from the features $\mathbf{x}_i$. Therefore, the difference between the individual contribution to the performance metric $G(y_i;\mathbf{x}_i;\delta_0)$ and the individual benchmark $\phi_{i,0}$ is explained by individual XPER values $\phi_{i,j}$.

As an illustration, consider the R^2 of a linear regression. Then, the individual XPER value $\phi_{i,j}$ can be expressed as

$$\phi_{i,j} = \sigma_y^{-2}\left(\hat{\beta}_j(x_{i,j} - \mathbb{E}(x_j))A - \hat{\beta}_j^2(x_{i,j}^2 - \mathbb{E}(x_j^2)) + \sum_{\substack{k=1\\k\neq j}}^{q} \hat{\beta}_k\hat{\beta}_j\sigma_{x_k,x_j}\right), \tag{12}$$

with $A = \left(2y_i - \sum_{\substack{k=1\\k\neq j}}^{q} \hat{\beta}_k(x_{i,k} + \mathbb{E}(x_k))\right)$. See Online Appendix K.2 for the proof. The value $\phi_{i,j}$ measures the contribution of feature x_j to $G(y_i;\mathbf{x}_i;\delta_0) = 1 - \sigma_y^{-2}(y_i - \hat{f}(\mathbf{x}_i))^2$. A positive individual XPER value $\phi_{i,j}$ means that incorporating the information included in feature x_j improves model prediction for individual i compared with the benchmark, hence increasing R^2. This gives rise to several observations. First, if $\hat{\beta}_j = 0$, that is, the feature has no impact on the model outcome, the feature x_j does not have any impact on the R^2, for all individuals. Second, the closer the realization of feature x_j is to its expected value $\mathbb{E}(x_j)$, the lower is the contribution of this feature to the performance metric, for all individuals. A similar result occurs when x_j^2 is close to its expected value. Indeed, when the characteristics of an individual are close to the mean values in the population, the contribution to the predictive performance of the model is also close to the average contribution of other individuals. We include a numerical example in Online Appendix C.2 to provide further intuition.

4.3.2. Dealing with Model Heterogeneity. Individual XPER values can be particularly useful to detect and understand sources of *model heterogeneity*. In machine learning and econometrics, *model homogeneity* refers to a consistent relationship between features and the target variable across the entire data set, implying that a single model can adequately capture the underlying patterns and make accurate predictions for all data points. Conversely, *model heterogeneity* arises when this relationship varies across different parts of the data set, indicating that a single model may fail to capture the diverse patterns within the data. In such cases, different models may be needed to achieve better predictive accuracy, indicating distinct feature-target relationships within subgroups of the data.

To identify these subgroups, we suggest relying on the individual XPER values. Indeed, by definition, individual XPER values measure the contribution of features on individual predictive performance, hence capturing the relationship between the features and the target variable. Intuitively, if a feature's impact on performance varies between two groups, it indicates that the feature does not exert the same influence on the accuracy of the classification or regression model.

In practice, model heterogeneity can be revealed by applying standard clustering techniques to individual XPER values, such as the K-medoids algorithm (Park and Jun 2009). Specifically, we propose the following three-step approach. First, using the original classification or regression model applied to the entire training sample, we calculate the individual XPER values based on the features $\mathbf{x}$ and the true target variable y. Second, we apply a clustering algorithm to the XPER values to identify C clusters that reveal the heterogeneity in the model's performance. Third, we estimate a separate model for each of the C subgroups derived from the initial sample. The empirical application presented in Section 7.3 suggests that this approach could lead to an improvement in the model's accuracy.

5. Estimation

In this section, we discuss the estimation procedure for the XPER values ϕ_j and $\phi_{i,j}$, based on a test sample $S_n = \{\mathbf{x}_i, y_i, \hat{f}(\mathbf{x}_i)\}_{i=1}^n$. Below, we assume that the model has a small number of features, typically $q < 10$.[5] When the number of features is larger, an approximation of the XPER values is required and we propose a modified version of the kernel SHAP method of Lundberg and Lee (2017) (see Online Appendix E.1 for more details).

Under Assumption 1, the XPER value ϕ_j can be estimated by a weighted average of individual contributions differences, such as

$$\hat{\phi}_j = \sum_{S\in\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})} w_S\left[\frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v;x_{v,j},\mathbf{x}_v^S,\mathbf{x}_u^{\bar{S}};\hat{\delta}_n) - \frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v;x_{u,j},\mathbf{x}_v^S,\mathbf{x}_u^{\bar{S}};\hat{\delta}_n)\right], \tag{13}$$

with S a coalition, that is, a subset of features, excluding the feature of interest x_j, and $\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})$ the powerset of the set $\{\mathbf{x}\}\setminus\{x_j\}$.

Table 2 illustrates the computation of the estimated value $\hat{\phi}_1$ associated with feature x_1 in a model with three features (x_1, x_2, x_3). For each coalition of features

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Table 2. Computation of the XPER Value $\hat{\phi}_1$ in a Three-Feature Model

S	w_S	$\frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{v,j}, \mathbf{x}_v^S, \mathbf{x}_u^{\bar{S}}; \hat{\delta}_n) - \frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{u,j}, \mathbf{x}_v^S, \mathbf{x}_u^{\bar{S}}; \hat{\delta}_n)$
$\{\emptyset\}$	1/3	$\frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{v,1}, x_{u,2}, x_{u,3}; \hat{\delta}_n) - \frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{u,1}, x_{u,2}, x_{u,3}; \hat{\delta}_n)$
$\{x_2\}$	1/6	$\frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{v,1}, x_{v,2}, x_{u,3}; \hat{\delta}_n) - \frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{u,1}, x_{v,2}, x_{u,3}; \hat{\delta}_n)$
$\{x_3\}$	1/6	$\frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{v,1}, x_{u,2}, x_{v,3}; \hat{\delta}_n) - \frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{u,1}, x_{u,2}, x_{v,3}; \hat{\delta}_n)$
$\{x_2, x_3\}$	1/3	$\frac{1}{n}\sum_{v=1}^{n} G(y_v; x_{v,1}, x_{v,2}, x_{v,3}; \hat{\delta}_n) - \frac{1}{n^2}\sum_{u=1}^{n}\sum_{v=1}^{n} G(y_v; x_{u,1}, x_{v,2}, x_{v,3}; \hat{\delta}_n)$

Note. This table presents information on the empirical XPER value computation, that is, the coalitions (column 1), the associated weights (column 2), and the estimated marginal contributions (column 3).

(column 1), we report the corresponding weight (column 2), along with the estimated marginal contribution of feature x_1 to the performance metric (column 3). The intuition is as follows: The sum over index u refers to the marginalization effect, whereas the sum over index v refers to the averaging effect. For a given instance v, we compute its average performance metric by replacing the features $x^{\bar{S}}$ that are *not included* in the coalition by the corresponding values observed for all the instances of the test sample (marginalization). Then, we compute the average performance metric for all the instances v (averaging).

Similarly to the estimation of features ϕ_j, we can estimate the *individual* XPER values $\phi_{i,j}$.[6] For any individual i of the test sample of S_n and any feature x_j, a consistent estimator of the individual XPER values $\phi_{i,j}$ is defined as

$$\hat{\phi}_{i,j} = \sum_{S\in\mathcal{P}(\{\mathbf{x}\}\setminus\{x_j\})} w_S \left[\frac{1}{n}\sum_{u=1}^{n} G(y_i; x_{i,j}, \mathbf{x}_i^S, \mathbf{x}_u^{\bar{S}}; \hat{\delta}_n) - \frac{1}{n}\sum_{u=1}^{n} G(y_i; x_{u,j}, \mathbf{x}_i^S, \mathbf{x}_u^{\bar{S}}; \hat{\delta}_n)\right]. \quad (14)$$

By definition, these individual XPER values satisfy

$$\hat{\phi}_j = \frac{1}{n}\sum_{i=1}^{n} \hat{\phi}_{i,j}. \quad (15)$$

6. Simulations

6.1. Baseline

In this section, we conduct a Monte Carlo simulation experiment to illustrate the XPER methodology and to highlight some of its properties. In this experiment, XPER values are used to explain the predictive performance of a probit model, measured by the AUC computed on a test set S_n. To understand the mechanisms of the XPER decomposition, we consider a white box model for which the predictions are explainable. The DGP is given by a latent variable model such that $y_i = \mathbf{1}(y_i^* > 0)$, where $\mathbf{1}(.)$ is the indicator function, $y_i^* = \boldsymbol{\omega}_i\beta + \varepsilon_i$, $\boldsymbol{\omega}_i = (1 : \mathbf{x}_i')$ and ε_i an independent and identically distributed (i.i.d.) error term with $\varepsilon_i \sim \mathcal{N}(0,1)$. We consider three i.i.d. features such that $\mathbf{x}_i = (x_{i,1}, x_{i,2}, x_{i,3})' \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ with $diag(\boldsymbol{\Sigma}) = (1.2, 1, 1)$. The true vector of parameters is $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)' = (0.05, 0.5, 0.5, 0)'$ with β_0 the intercept.

We simulate $K = 5{,}000$ pseudo-samples $\{y_i^k, \mathbf{x}_i^k\}_{i=1}^{T+n}$ of size 1,000, for $k = 1, \ldots, K$. For each pseudo-sample, we use the first $T = 700$ observations to estimate a probit model and the remaining ones as test set S_n to compute the AUC and the corresponding XPER values according to Equation (13). For instance, the estimated parameters obtained for the simulation $k = 1$ are equal to $\{\hat{\beta}_0^k, \hat{\beta}_1^k, \hat{\beta}_2^k, \hat{\beta}_3^k\} = \{0.0109, 0.4943, 0.5234, 0.0688\}$ and the AUC^k is equal to 0.7775. For this simulation, the average feature contributions, taken over the individuals of the test set, are the following:

$$\underbrace{0.7775}_{AUC^k} = \underbrace{0.4984}_{\hat{\phi}_0^k} + \underbrace{0.1716}_{\hat{\phi}_1^k} + \underbrace{0.1098}_{\hat{\phi}_2^k} + \underbrace{(-0.0023)}_{\hat{\phi}_3^k}.$$

As expected, the estimated benchmark $\hat{\phi}_0^k$ is close to 0.5. As a reminder, an AUC equal to 0.5 is associated with a random predictor. Hence, the benchmark $\hat{\phi}_0^k$ corresponds to the AUC of the model that we would obtain on a virtual test sample in which the target variable is independent from all features. The difference between the estimated AUC and this hypothetical benchmark is explained by the feature contributions. The feature with the largest variance (x_1) has also the largest contribution to the predictive ability of the model ($0.1716/(0.7775 - 0.4984) \simeq 62\%$). In contrast, the contribution of x_3, which is not used to generate the data ($\beta_3 = 0$), is very close to zero.

Figure 1 displays the empirical distributions of the AUC, the benchmark values $\hat{\phi}_0$, and the XPER values associated with features x_1, x_2, and x_3, computed from the K simulations. It confirms the robustness of our analysis, but also illustrates the possibility to make inference on XPER values by Bootstrap or other numerical methods.

In Table 3, we report the values of the individual XPER values obtained for the first simulation $k = 1$. The third column displays the individual benchmark. For a binary classification model, the benchmark $\hat{\phi}_{i,0} \equiv \hat{\phi}_{i,0}(y_i)$ only takes two values. In our simulations, for individuals $y_i = 1$ (respectively, $y_i = 0$), this value is equal to 0.5001 (respectively, 0.4967). Remember that the individual benchmark corresponds to the contribution to the AUC obtained from a random predictor for an individual with a target value $y_i = 1$ or

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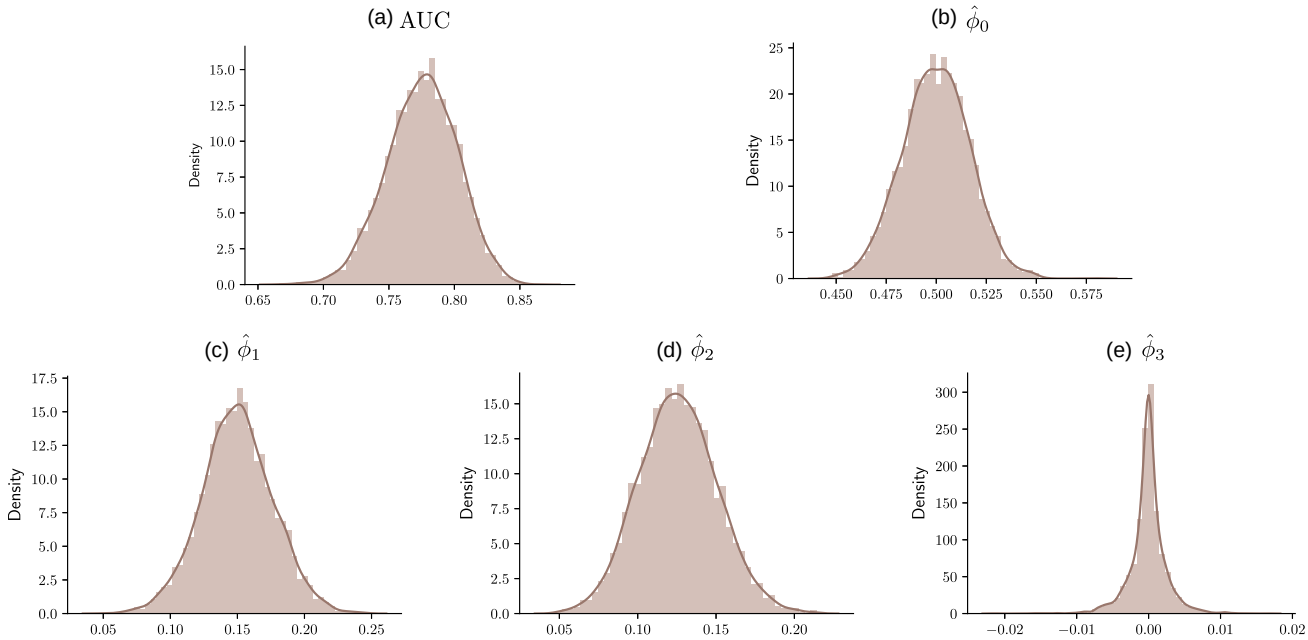
Figure 1. (Color online) Empirical Distributions of AUC and XPER Values



Notes. This figure displays the empirical distributions of the AUC and XPER values on the test sample according to the DGP detailed in Section 6.1. The solid lines refer to kernel density estimations.

$y_i = 0$. Consider the first instance $i = 1$ with $y_i^k = 1$, $G(y_1^k; \mathbf{x}_1^k; \hat{\delta}_n^k) = 0.9$ and $\hat{\phi}_{1,0} = 0.5001$. As $G(y_1^k; \mathbf{x}_1^k; \hat{\delta}_n^k) > \hat{\phi}_{1,0}^k$, it means that the features allow the probit model to better predict the event y_1 for this instance than a random predictor.

Finally, columns 4, 5, and 6 report the individual XPER values associated with features x_1, x_2, and x_3. First, we verify that feature x_3 has close to no impact on the AUC for all instances. Second, the heterogeneity of contributions to the AUC depends on individual characteristics. Remember that at the global level, the contribution of feature x_1 is higher than the one of feature x_2. However, at the local level, we observe for $i = 300$ that the contribution of feature x_1 is smaller than the one of feature x_2, that is, $0.1316 < 0.3069$. Third, some individual contributions are negative. For instance, the negative contribution of feature x_2 for individual $i = 297$ implies that, for this instance, this feature hinders the model from predicting the true target value $y_i^k = 0$.

To further illustrate the XPER methodology, we provide in Online Appendix F two additional Monte

Table 3. Illustration of the AUC XPER Decomposition for Simulation $k = 1$

Panel A: Local analysis (individual XPER values)							
	$G(y_i^k; \mathbf{x}_i^k; \hat{\delta}_n^k)$	$\hat{\phi}_{i,0}^k$	$\hat{\phi}_{i,1}^k$	$\hat{\phi}_{i,2}^k$	$\hat{\phi}_{i,3}^k$	y_i^k	$\hat{p}_i^k$
$i = 1$	0.9000	0.5003	0.2449	0.1770	−0.0222	1	0.6614
$i = 2$	1.0000	0.5003	0.3975	0.1167	−0.0145	1	0.8369
…	…	…	…	…	…	…	…
$i = 297$	0.2533	0.4966	−0.1232	−0.1237	0.0035	0	0.7616
…	…	…	…	…	…	…	…
$i = 300$	0.9600	0.4966	0.1316	0.3069	0.0249	0	0.1982
Panel B: Global analysis (XPER values)							
	AUC^k	$\hat{\phi}_0^k$	$\hat{\phi}_1^k$	$\hat{\phi}_2^k$	$\hat{\phi}_3^k$	$\frac{1}{n}\sum_{i=1}^n y_i^k$	$\frac{1}{n}\sum_{i=1}^n \hat{p}_i^k$
	0.7775	0.4985	0.1716	0.1098	−0.0024	0.4967	0.4941

Notes. This table presents both the individual XPER values (local analysis, Panel A) and the aggregated XPER values (global analysis, Panel B) associated with the AUC of a Probit model for each of the three features x_j, where $j = 1, 2, 3$. The values correspond to those obtained for the test sample (sample size $n = 300$) during the first simulation ($k = 1$). The additional columns provide the individual contributions to the AUC, the individual benchmarks, the target variable, and the conditional probabilities $\hat{p}_i^k = \hat{\mathbb{P}}(y_i^k = 1 | \mathbf{x}_i^k)$ derived from the Probit model.

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Carlo simulation experiments illustrating how XPER values can be used to detect the origin of overfitting.

6.2. Stability

Statistical stability is a legitimate concern for algorithmic explainability methods (Visani et al. 2021, Chen et al. 2024). When estimating the XPER values, three main sources of noise can arise, each of which may potentially affect the statistical stability of the XPER decomposition: (a) the estimation of the model on the training data set, (b) the estimation of the performance metric on the test data set, and (c) the approximation of the XPER value by the kernel-SHAP approach. In this section, we study the effect of source (b) on the XPER decomposition using a simulation study.[7] We consider a DGP defined by a regression model with nonlinear effects, including quadratic terms, a threshold effect, and an interaction term:

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,2}^2 + \beta_4 \mathbf{1}_{(x_{i,1}>0.4)} + \beta_5 x_{i,1} x_{i,2} + \varepsilon_i, \quad i = 1, \ldots, T+n, \tag{16}$$

where ε_i is an i.i.d. error term, with $\varepsilon_i \sim \mathcal{N}(0,1)$, T and n are the training and test sample sizes, and $\mathbf{1}(.)$ refers to the indicator function.

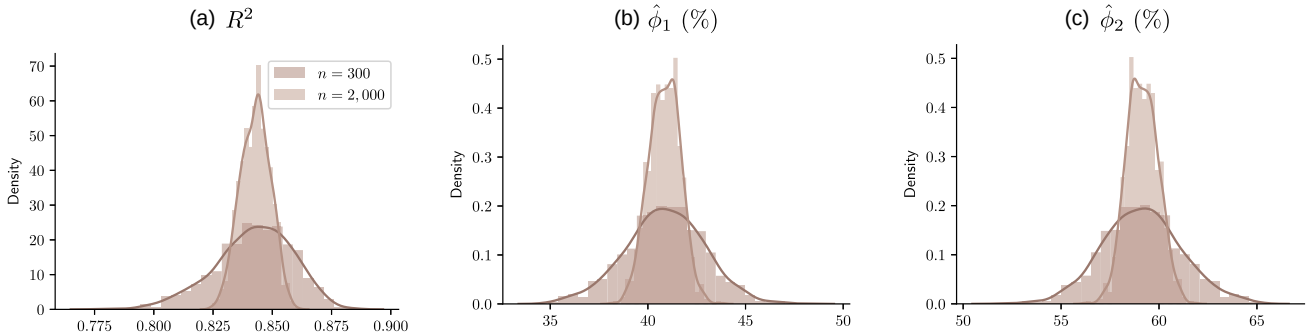
For clarity, we distinguish between the *primary variables* $(x_{i,1}, x_{i,2})$ and the derived *features* included in the model specification, namely $(x_{i,1}, x_{i,2}, x_{i,2}^2, \mathbf{1}(x_{i,1} > 0.4), x_{i,1}x_{i,2})$. The primary variables are drawn from Beta distributions, with $x_{i,1} \sim \text{Beta}(2,5)$ and $x_{i,2} \sim \text{Beta}(3,2)$. Moreover, we introduce dependence between these two variables using a Gaussian copula with zero mean and correlation matrix $\mathbf{\Sigma}$, defined as $\mathbf{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. The parameter vector is defined as $(\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)' = (0.05, 5, 5, 2.5, 1, 1)'$, and the correlation coefficient ρ is set to 0.3.

In this experiment, we keep the model fixed and vary only the *test data set* to assess the variability of the XPER values caused by source (b). This corresponds to analyzing how the finite size of the test sample affects both the performance metric and the XPER estimates. To isolate this specific source of variability and avoid confounding effects, we design the simulation study as follows. The model is estimated only once on a training sample $\{y_i, \mathbf{x}_i\}_{i=1}^T$ of size $T = 700$, thereby mitigating source (a). Moreover, we compute the exact XPER values by considering all possible feature coalitions, which eliminates source (c) associated with the kernel-based method. We then generate $S = 1{,}000$ independent test samples $\{y_i^s, \mathbf{x}_i^s\}_{i=1}^n$ of size $n = 300$ and 2,000. For each test sample s, we evaluate the model's predictive performance using the R^2 and compute the corresponding XPER values.

Figure 2 displays the empirical distributions of R^2 and XPER values associated with the primary variables. This figure highlights two key findings. First, increasing the test sample size reduces the variability of both the empirical R^2 and the corresponding XPER estimates. This result follows directly from the weak law of large numbers: As the test sample size tends to infinity, the empirical means used to estimate the feature contributions within each coalition (see Equation (13)) converge in probability to their theoretical population counterparts (Definition 3) used to define the XPER values. For instance, when $n = 300$, the XPER value ϕ_1 ranges from 35% to 47.5%, whereas for $n = 2{,}000$, its range narrows considerably, from 38.5% to 43%.

Second, even with a relatively small sample size of $n = 300$, the XPER values remain relatively stable, as the variability induced by source (b) does not appear to affect the qualitative interpretation of the model's performance decomposition. In all simulated test samples, the estimated XPER value for the second primary variable consistently exceeds that of the first. Thus, although the XPER values fluctuate slightly across test samples, this variability does not alter the economic interpretation of the results. Obviously, this conclusion depends on several factors, including the test sample size, the model, and the features. However,

Figure 2. (Color online) Empirical Distributions of R^2 and XPER Values Associated to Primary Variables



Notes. This figure displays the empirical distributions of the R^2 and XPER values for the two primary variables x_1 and x_2. These values are estimated for $S = 1{,}000$ pseudo test samples of size $n = 300$ and $n = 2{,}000$. The solid lines refer to kernel density estimates.

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this source of instability is inherent to any empirical evaluation based on a finite test sample and is not different from the uncertainty one faces when interpreting the R^2 or any other performance metric.

7. Empirical Application

7.1. Data and Model

We implement our methodology on a proprietary database of auto loans provided by an international bank. For each borrower, information is available on whether a default has occurred ($y = 1$) on the loan or not ($y = 0$). Given the sensitive nature of the data, we had to randomly under-sample individuals to set the default rate to an arbitrary 20% level. Besides benefits in terms of confidentiality, setting a high arbitrary default rate also protects us against concerns arising from using an unbalanced database. After undersampling, our database includes 7,440 borrowers.[8] Besides the default target variable, we have access to 10 features on the loan (funding amount, funded duration, vehicle price, down payment) and on the borrower (job tenure, age, marital status, monthly payment in percentage of income, home ownership status, credit event). We divide the database into a stratified training (70%) and test (30%) samples to have the same default rate in both subsamples.

We provide in Table A5 of Online Appendix H.1 some summary statistics about the features and the target variable. In our data set, a typical loan amounts to around 11,500 euros, finances a 13,000 euro car, and lasts for 56 months. A typical borrower is 45 years old, married, not being home-owner, has spent nine years in the same job, experienced no credit event over the past six months, provides less than a 50% down payment, and allocates 10% of monthly income to reimburse the car loan. To get a first sense of the role of each feature on default, we display in Figure A8 in the online appendix their distributions separately for defaulting and nondefaulting borrowers. This preliminary test indicates that the list of discriminating feature includes age, credit event, down payment, marital status, and ownership status.

Using the training sample, we estimate an XGBoost model to predict default. We selected this model because it is recognized as one of the most powerful scoring engines (Gunnarsson et al. 2021). Another reason for using an XGBoost is its black box nature. Indeed, although the XPER methodology is model-agnostic, we believe it is interesting to assess its usefulness when used with a particularly complex and opaque algorithm. We select the hyperparameters of the XGBoost using stratified fivefold cross-validation based on a balanced accuracy criterion (Bergstra and Bengio 2012).

Table 4 displays the values of six performance metrics for the XGBoost model obtained on the training and test samples. Specifically, (1) the AUC measures the discriminatory ability of the model, (2) the Brier score evaluates the accuracy of the probabilities, and (3) accuracy, balanced accuracy (BA), sensitivity, and specificity assess the correctness of the categorical predictions. As shown in Table 4, the XGBoost has an AUC of 0.7521, a Brier score of 0.1433, and an accuracy of 79.53 on the test sample. Despite the cross-validation of the hyperparameters, we observe some overfitting in the model as its performances drops slightly from the training sample to the test sample. Overall, the performance metrics of our model are comparable to those reported in Gunnarsson et al. (2021).

7.2. XPER Decomposition

We display in Figure 3(a) the decomposition of the AUC among the 10 features obtained with the estimation method detailed in Online Appendix E.1.[9] For ease of presentation, we express the feature contributions in percentage of the spread between the AUC and its benchmark (see Equation (5)). As shown in Table 4, the AUC in the test sample is equal to 0.7521, which is significantly better than the benchmark value of 0.5 obtained for a random predictor.[10] We see that around 40% of this overperformance is coming from *funding amount*. The second most contributing feature is *job tenure*, which accounts for another 18%. It is interesting to note that with only two features, we can explain more than half of the performance of the model. Next, we have five features which contribute each for another 8%–10% to the performance. At the other side of the spectrum, *down payment* is the feature contributing the least to the AUC. On the test sample, this feature does not help the model to better predict default than a random predictor as its XPER value is close to zero. Note that in Online Appendix H.4, we also analyze the effect of the various features on the performance for each borrower individually.

We now compare the XPER performance decomposition to standard feature contribution methods commonly

Table 4. Model Performance

Sample	Size (%)	AUC	Brier score	Accuracy	BA	Sensitivity	Specificity
Training	70	0.8969	0.0958	86.98	72.43	48.18	96.69
Test	30	0.7521	0.1433	79.53	58.69	23.99	93.39

Notes. This table displays the performance metrics of the XGBoost model on the training and test samples. BA, balanced accuracy.

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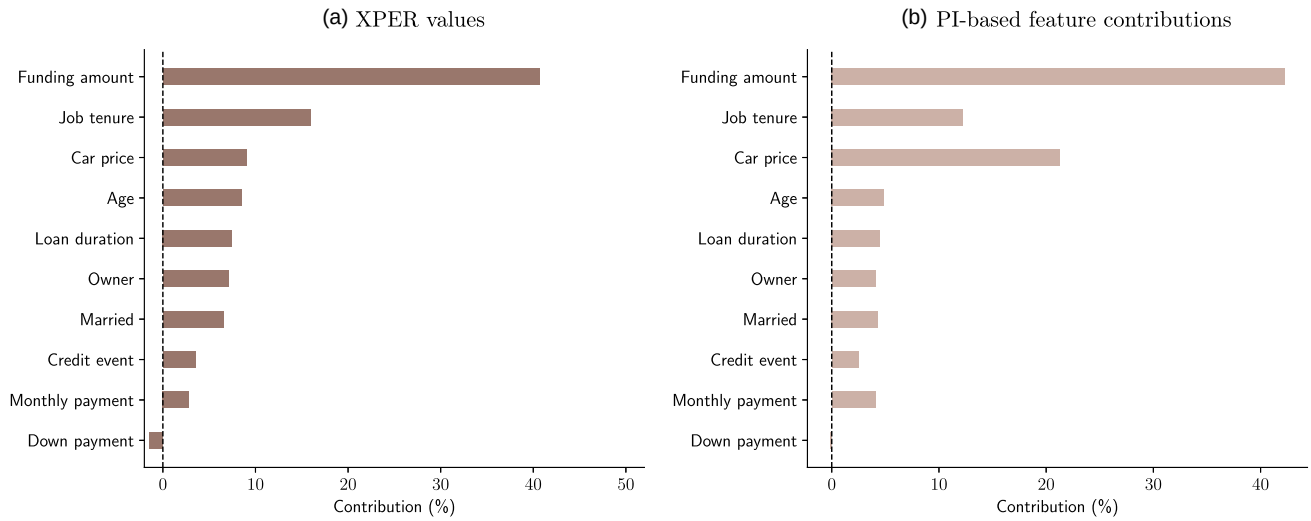
Figure 3. (Color online) XPER Decomposition and Permutation Importance



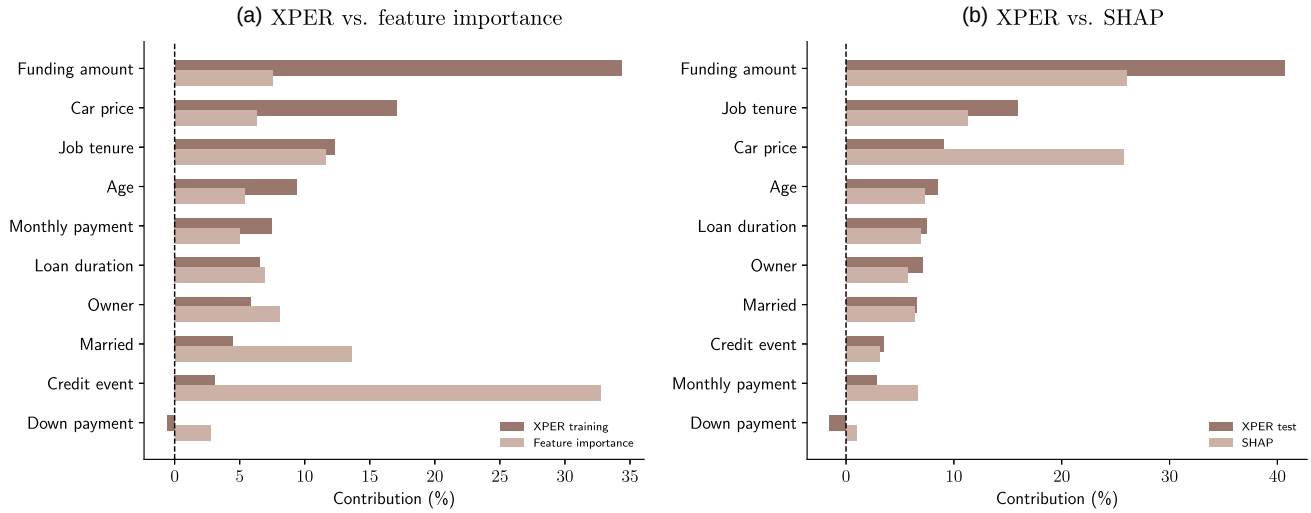
Note. This figure displays in Panel (a) the XPER values for the AUC of the XGBoost model estimated on the test sample and in Panel (b) the feature contributions for the AUC based on Permutation Importance (PI).

used in machine learning to assess the impact of a feature on performance or to explain the output of a black box model, namely permutation importance (PI), feature importance, and SHAP values.

First, we distinguish between the XPER decomposition of the AUC and the PI method introduced by Breiman (2001). The latter computes for a feature x_j, the decrease in the AUC of the model when the values of this feature are randomly reshuffled across instances. As the permutation breaks the dependency between the target variable and the feature, the resulting drop in model AUC indicates how much the model depends on the feature. In our analysis, the PI results indicate the average decrease in AUC when the values are reshuffled 200 times to obtain more robust results. As for the previous methods, we divide each feature contribution by the sum of all features contributions to compare them to the XPER values. We see in Figure 3(b) that XPER and PI produce different results. For instance, the contribution of the car price is twice as important with PI than with XPER.[11]

Second, we contrast in Figure 4(a) the XPER values and the XGBoost-based feature importance. The latter computes for a feature x_j, the average gain across all splits where the feature is utilized. For ease of comparison, we divide each feature contribution by the sum of the ten feature contributions. As shown in Figure 4(a), the result is rather striking. Indeed, the two methodologies lead to very different contributions as some dominating features in a given

Figure 4. (Color online) XPER vs. Other Feature Contribution Methodologies



Note. This figure compares the XPER values of the AUC with (a) the XGBoost-based feature importances and (b) the average absolute SHAP values.

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methodology play a minor role in the other. For instance, *credit event* exhibits the highest feature importance but it is only the nineth most contributing feature according to XPER. Differently, *funding amount* plays a very important role to explain performance but does not contribute much in terms of feature importance.

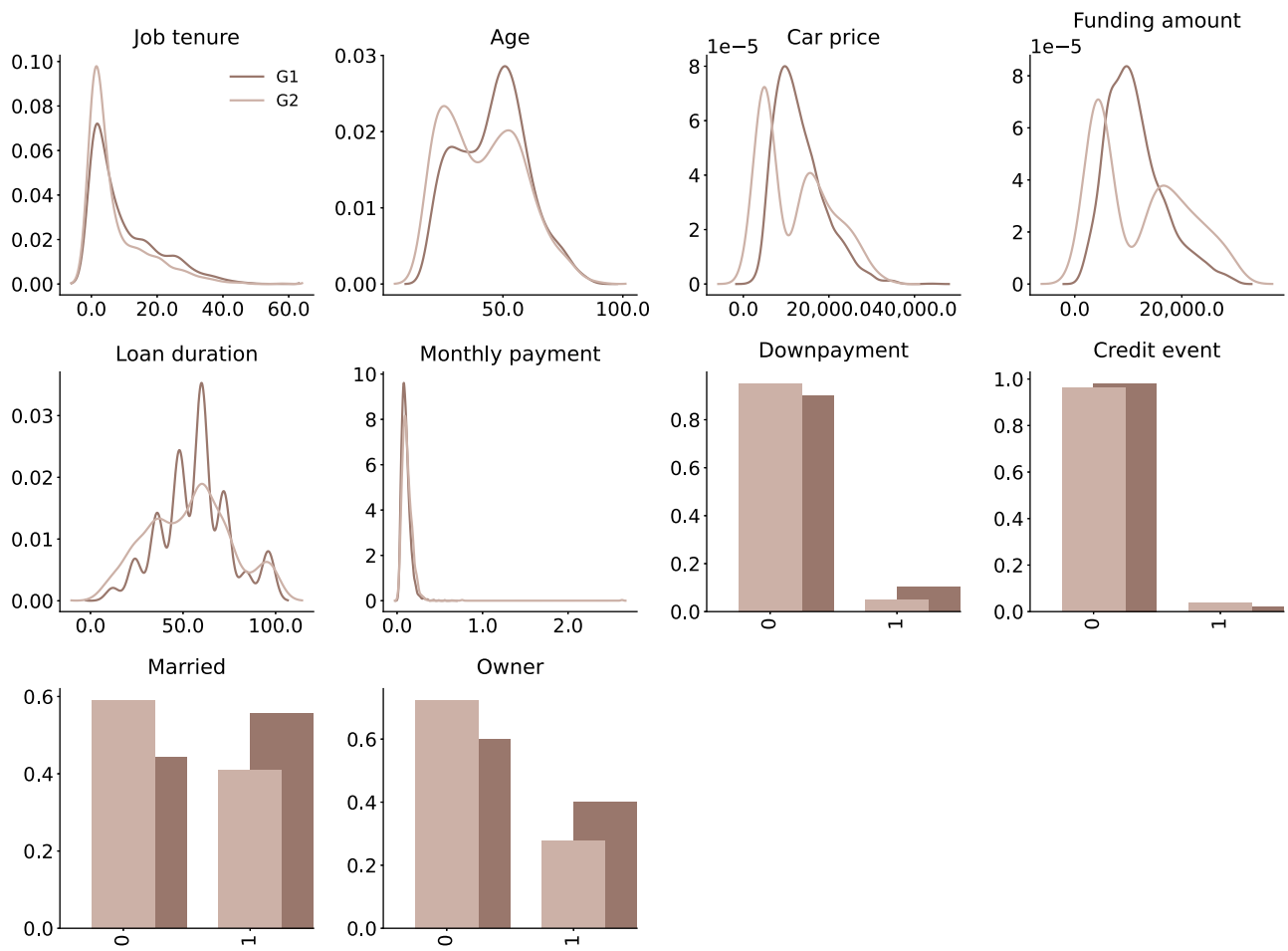
Third, we compare in Figure 4(b) the XPER decomposition of the AUC with the SHAP values introduced by Lundberg and Lee (2017). In our context, the SHAP values assess the impact of the different features on the probabilities of default of each borrower. As it is the norm, we take the average absolute SHAP values for each feature to assess the feature contribution at the model level. As before, we divide each feature contribution by the sum of all features contributions to express them in percentages. In Figure 4(b), we see that SHAP and XPER provide for some features very different information. For instance, the *car price* contribution is more than twice as important for SHAP than for XPER. Similarly, the *funding amount* contribution is around 40% for XPER, whereas only 28% for SHAP.[12]

7.3. Using XPER to Boost Model Performance

We now show how XPER can be used to deal with heterogeneity issues and improve performance. In practice, it is often challenging to estimate a single model able to correctly estimate the relationship between the target variable and the features for all individuals in the sample. In this section, we propose an alternative two-step procedure. First, we build homogeneous groups of individuals displaying similar XPER values using a clustering algorithm. Second, within each group, we estimate a group-specific model. Given the definition of XPER, in each group, the features have a similar impact on the target variable. This alternative approach is likely to increase the model performance compared with the one-fits-all model, which has been used thus far in the empirical application.

In the first step, we estimate the XPER values $\hat{\phi}_{i,j}$ on the training sample and apply the K-medoids method (Park and Jun 2009) to create two clusters of individuals with similar XPER values.[13] We see in Figure 5 that individuals in group 1 tend to be older, married,

Figure 5. (Color online) Features Distribution by Group Based on XPER Values



Notes. This figure displays the distribution of the features on the training sample by group created from individual XPER values using the K-medoids methodology. For continuous features, we use kernel density estimation. Dark gray refers to the first group and light gray to the second group.

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homeowners, and ask for a midrange loan to finance a midpriced car. One striking difference is that in group 2, individuals tend to finance cars with extremely low prices or high prices. Overall, group 1 gathers individuals displaying a lower risk profile compared with those in group 2. This is confirmed by the fact that the default rate is much higher in group 2 (52%) than in group 1 (13%). Moreover, as shown in Figure A13, a and b, in Online Appendix H.5, the feature distributions are very different for defaulters and nondefaulters in each group. We also see in Figure A14 in the online appendix that, in group 2, using the car price deteriorates the performance of the model because its XPER value is close to −20% versus 20% in group 1. Therefore, these results suggest that estimating group-specific models is likely to boost predictive performance.

In the second step, we estimate using the training data set one XGBoost model per group to predict default. Then, we aim to compare the resulting out-of-sample performance with the one of the one-fits-all model. To do so, we first assign each individual of the test sample to either group 1 or group 2 using the clustering rule built on the training sample. We see in Table 5 that the performance of our strategy (column 2) is significantly higher than the performance of the initial model (column 1). For instance, the AUC of the model on the test sample increases by 16 percentage points, from 0.752 to 0.912. Another notable result is that by using two distinct models, the bank would be able to correctly detect 64% of the defaulting borrowers against only 24% with the initial model. Yet, it would still be able to correctly detect more than 95% of the nondefaulting borrowers.

Finally, we benchmark our strategy with a standard clustering approach based on the features themselves. Specifically, we use the K-prototype algorithm (Huang 1997) to create two clusters of individuals with comparable feature values.[14] As shown in column 3, the performance of the standard approach is close to the one of the one-fits-all model (AUC = 0.744 versus 0.752) and is much lower than the performance of the XPER-based models.

Table 5. Model Performance

	Initial (1)	Clusters on XPER values (2)	Clusters on features (3)
AUC	0.752	0.912	0.744
Brier score	0.143	0.080	0.151
Accuracy	79.53	89.11	79.53
Balanced accuracy	58.69	79.74	59.11
Sensitivity	23.99	64.13	25.11
Specificity	93.39	95.35	93.11

Notes. This table displays the performances on the test sample of (1) the initial XGBoost model, (2) our two-step strategy relying on the clustering of XPER values, and (3) the alternative two-step strategy based on the clustering of feature values. For each clustering-based approach, two clusters were derived using the training sample. The performances reported in columns (2) and (3) represent the average performance metrics across the two clusters on the test sample.

8. Conclusion

We have introduced XPER, a methodology designed to measure the feature contributions to the performance of any regression or classification model. XPER is built on Shapley values and interpretability tools developed in machine learning but with the distinct objective of focusing on model performance, such as AUC or R^2, and not on model predictions, $\hat{y}$.

Specifically, XPER breaks down the difference between a performance metric of the model and a benchmark value. The latter corresponds to the performance metric that we would obtain on a hypothetical sample in which the target variable is independent from all the features included in the model. As such, the XPER decomposition only focuses on the part of the performance that directly originates from the features. Other advantages of the method include being implementable either at the model level or at the individual level and not being plagued by omitted variable biases, because it does not require re-estimating the model with different features coalitions.

We show that identifying the driving forces of the performance of a predictive model is very useful in practice. In a loan default forecasting application, XPER appears to be able to efficiently deal with heterogeneity issues and to boost out-of-sample performance. To do so, we create homogeneous groups of borrowers by clustering them based on their individual XPER values. We find that estimating group-specific models yields to a higher predictive accuracy than with a one-fits-all model.

XPER could go beyond performance. Indeed, as XPER can decompose any function of both $\hat{y}$ and y, one could use it to identify the features responsible for the lack of algorithmic fairness of a given machine learning model (Hurlin et al. 2026). Another interesting avenue for future research relies on developing inference methods for XPER values, enabling the assessment of their statistical significance.

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seminar participants at Aix-Marseille School of Economics, Amundi, AXA Investment Managers, COFACE, and University of Orléans for comments.

Endnotes

[1] *Efficiency* guarantees that the total contribution of all features equals the overall model performance metric, minus the benchmark value. *Symmetry* ensures that features making equivalent contributions receive identical XPER values. *Linearity* stipulates that if model performance can be decomposed into separate parts, the XPER value of the whole model equals the sum of its parts. The *null effect* axiom ensures features that do not influence model performance receive a XPER value of zero.

[2] An alternative approach would be to compute the expectation over the conditional distribution of the excluded features, $\mathbf{x}^{\bar{S}}$, given the joint distribution of the target variable y, the feature of interest x_j, and the features included in the coalition $\mathbf{x}^S$, i.e., $\mathbb{E}_{\mathbf{x}^{\bar{S}}}|y, x_j, \mathbf{x}^S$. However, the corresponding decomposition would no longer satisfy the four axioms mentioned in Section 4.2. See the discussion about the conditional expectation Shapley (CES) methods in Sundararajan and Najmi (2020).

[3] Note that the XPER values ϕ_j can be negative, even if the DGP of the *training* and *test* samples are identical. This occurs in a particular case where the true parameter β_j of the feature of interest x_j is null. In this case, the parameter estimate $\hat{\beta}_j$ obtained using the *training sample* can be either positive or negative. Additionally, the estimate of the covariance between the target variable y and the feature x_j, denoted $\hat{\sigma}_{y,x_j}$, obtained using the *test sample* can also be either positive or negative, even if the true covariance σ_{y,x_j} is null as $\beta_j = 0$. Therefore, if the sign of the parameter estimate $\hat{\beta}_j$ and the covariance estimate $\hat{\sigma}_{y,x_j}$ differ, then the XPER value ϕ_j can be negative.

[4] Online Appendix C.1 presents a detailed example illustrating the interpretation of the benchmark value ϕ_0 in the context of a standard linear regression model and the decomposition of the R^2.

[5] The standard estimation framework is only feasible for a small number of features q. The computation of the XPER values ϕ_j according to Definition 3 becomes cumbersome as the number of features increases. Indeed, it requires to consider $q \times 2^{q-1}$ coalitions of features, for example, 5,120 for $q = 10$ and 10,485,760 for $q = 20$, and so on.

[6] Performing inference on XPER values presents significant challenges. Williamson and Feng (2020) proposed an approach for statistical inference on a variable importance measure called the "Shapley population variable importance measure" (SPVIM). However, this approach is incompatible with XPER values due to its reliance on submodel re-estimation and its focus on global rather than individual analyses. Another potential approach involves using bootstrap methods to generate a block bootstrap distribution of XPER values, although the high computational cost remains an important limitation. Finally, conformal prediction (Vovk et al. 2005, Romano et al. 2019) represents a promising avenue, as it could be adapted to construct prediction intervals for individual XPER values.

[7] The impacts of noise sources (a) and (c) are investigated in Online Appendices G and E.2, respectively.

[8] Besides undersampling, potential techniques to deal with imbalanced data sets include oversampling (e.g., ADASYN, ROSE, SMOTE) and cost-sensitive learning. For more details, see Jiang et al. (2023) and Chen et al. (2024).

[9] See Online Appendix H.2 for the decomposition of several other performance metrics.

[10] In Online Appendix H.3, we vary the average default rate in the data set from 5% to 30% by randomly removing some default observations or nondefault observations from the sample, respectively, to assess the impact of imbalanced data on XPER values.

[11] For a more detailed comparison between XPER and PI, see Online Appendix I. We show that PI is a special case of XPER and is only designed for a global analysis.

[12] For a more detailed comparison between XPER and SHAP, see Online Appendix J.

[13] XPER values are always continuous, even when they are associated with categorical variables. To determine the optimal number of clusters, we used the "silhouette score" criterion. We report this score in Table A7 in Online Appendix H.5 for a number of clusters ranging from 2 to 10.

[14] We utilize the K-prototype algorithm instead of the K-medoids algorithm because the former is specifically designed to handle both categorical and continuous features. More precisely, when handling continuous features, the algorithm calculates the squared Euclidean distance between the features and a representative vector (or prototype) of clusters. For categorical features, it determines the number of mismatches between the features and the cluster's prototypes.

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